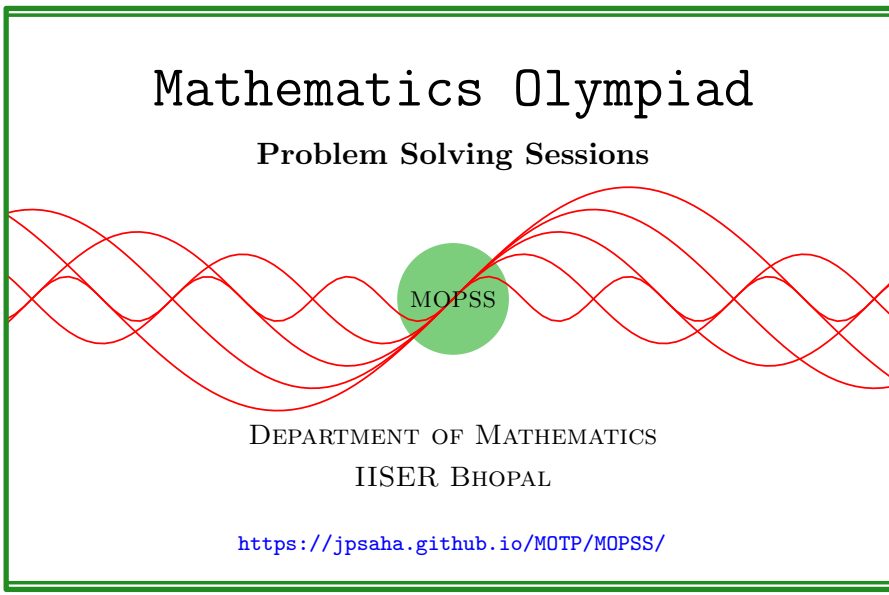
MOPSS

14 March 2026



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

Exercise 1 (Argentina National Olympiad 2018 Level 2 P1,). A list of 2018 numbers is created using the following procedure: the first number is 47, the second number is 74, and from there, each number is equal to the number formed by the last two digits of the sum of the two previous numbers:

$$47, 74, 21, 95, 16, 11, \ldots$$

Bruno squares each of the 2018 numbers and sums them. Determine the remainder when this sum is divided by 8.

Walkthrough —

(a) Consider the residues of the numbers in the list modulo 4.

(b) Determine the residues of the squares of the numbers in the list modulo 8.

(c) Determine the remainder when the sum of the squares of the numbers in the list is divided by 8.

Solution 1. Note that the first two integers on the list are congruent to 3 and 2 modulo 4, respectively. Therefore, the next six integers on the list are congruent to 1, 3, 0, 3, 3, and 2 modulo 4, respectively. Applying induction, it follows that the residues of the numbers in the list modulo 4 are periodic with period 6:

$$3, 2, 1, 3, 0, 3, 3, 2, 1, 3, 0, 3, \ldots.$$

Hence, the residues of the squares of the numbers in the list modulo 8 are also periodic with period 6:

$$1, 4, 1, 1, 0, 1, 1, 4, 1, 1, 0, 1, \ldots.$$

Since $2018 = 6 \cdot 336 + 2$, the sum of the squares of the numbers in the list is congruent to

$$336 \cdot (1 + 4 + 1 + 1 + 0 + 1) + 1 + 4$$

modulo 8. Therefore, the remainder when this sum is divided by 8 is 5. ■

Exercise 2 (All-Russian Mathematical Olympiad 2025 Grade 9 Day 2 P7, , proposed by I. A. Efremov). The numbers $1, 2, 3, \ldots, 60$ are written in a row in that exact order. Igor and Ruslan take turns inserting the signs $+, -, \times$ between them, starting with Igor. Each turn consists of placing one sign. Once all signs are placed, the value of the resulting expression is computed. If the value is divisible by 3, Igor wins; otherwise, Ruslan wins. Which player has a winning strategy regardless of the opponent's moves?

Walkthrough —

(a) Igor, the first player, has a winning strategy.

(b) Note that the numbers $1, 2, 3, \ldots, 30$ are congruent to $31, 32, 33, \ldots, 60$ modulo 3, respectively.

(c) Igor in his first turn places a $-$ sign between 30 and 31.

(d) After that, the numbers $1, 2, 3, \ldots, 30$ can be thought as grouped together and the numbers $31, 32, 33, \ldots, 60$ can also be thought as grouped together.

(e) For each choice of Ruslan, Igor can respond by placing the corresponding sign in the other half.

Solution 2. We claim that Igor, the first player, has a winning strategy. Note that the numbers $1, 2, 3, \ldots, 30$ are congruent to $31, 32, 33, \ldots, 60$ modulo 3, respectively. Igor in his first turn places a $-$ sign between 30 and 31. Next, after each turn of Ruslan, which is to place a sign in one of the two halves, that is, to place a sign between two consecutive numbers between 1 and 30 or to place a sign between two consecutive numbers between 31 and 60, Igor can respond by placing the corresponding sign in the other half. Here the corresponding sign means that if Ruslan places a $\times$ sign, then Igor also places a $\times$ sign, and if Ruslan places a $+$ sign, then Igor places a $-$ sign, and if Ruslan places a $-$ sign, then Igor places a $+$ sign. After all signs are placed, the resulting expression can be written as the difference of two expressions, one of which consists of the numbers $1, 2, 3, \ldots, 30$ and the other of which consists of the numbers $31, 32, 33, \ldots, 60$. The strategy of Igor ensures that these two expressions are congruent modulo 3. Therefore, Igor wins. ■

Exercise 3 (Moldova National Olympiad 2006 Grade 11 P4, , cf. QEDMO 2006 P11)**.** On a table, there are 2006 cards, each with a natural number written on it. Two players play a game, taking turns. In each turn, a player takes a card from either the leftmost or the rightmost end of the row of cards. The game ends when all cards are taken. After the game, each player calculates the sum of the numbers on the cards they have taken. If the sum of the first player is not less than the sum of the second player, then the first player wins. Show that the first player has a winning strategy. (Note that the integers on the cards are visible to both players throughout the game.)

Walkthrough —

(a) Show that the first player can always take the cards in the odd positions or the cards in the even positions.

(b) Before the game starts, the first player compares the sum of the numbers

> on the cards in the odd positions and the sum of the numbers on the cards in the even positions.

Solution 3. The first player can use the following strategy to win the game. Before the game starts, the first player calculates the sum of the numbers on the cards in the odd positions, and the sum of the numbers on the cards in the even positions. Since there are 2006 cards, one of these sums is greater than or equal to the other. If the sum of the numbers on the cards in the odd positions is greater than or equal to the sum of the numbers on the cards in the even positions, then the first player always takes the cards in the odd positions. Otherwise, the first player always takes the cards in the even positions. In either case, the first player wins.

Note that the first player can always take the cards in the odd positions or the cards in the even positions, because the first player starts the game and there are an even number of cards. Indeed, if the first player takes the card in the odd (resp. even) position in the first turn from the left (resp. right) end of the row, then after each turn of the second player, the first player can respond by taking the card adjacent to the card taken by the second player, which is in the odd (resp. even) position. Thus, the first player can always take the cards in the odd positions or the cards in the even positions, and hence ensures that the sum of the numbers on the cards taken by the first player is greater than or equal to the sum of the numbers on the cards taken by the second player. Consequently, the first player wins. ■

Exercise 4 (Taiwan TST 2024 Round 1 Quiz 1, , proposed by Ho-Chien Chen)**.** Let $n \geq 5$ be a positive integer. There are n stars with values 1 to n, respectively. Anya and Becky play a game. Before the game starts, Anya places the n stars in a row in whatever order she wishes. Then, starting from Becky, each player takes the left-most or right-most star in the row. After all the stars have been taken, the player with the highest total value of stars wins; if their total values are the same, then the game ends in a draw. Find all n such that Becky has a winning strategy.

> **Walkthrough** —
>
> **(a)**

Solution 4. We claim that Becky has a winning strategy if and only if $n \equiv 2 \pmod 4$. If $n \equiv 2 \pmod 4$, then compares the sum of the values of the stars in the odd positions, and the sum of the values of the stars in the even positions. Since $n \equiv 2 \pmod 4$, one of these sums is strictly greater than the other. If the sum of the values of the stars in the odd positions is greater than the sum of the values of the stars in the even positions, then Becky always takes the stars in the odd positions. Otherwise, Becky always takes the stars in the even positions. In either case, Becky wins.

If $n = 2m+1$ with $m \geq 2$, then Anya places the stars with values $1, 2, \ldots, m+1$ in the odd positions from left to right, and places the remaining stars in the even positions from left to right. Then, no matter how Becky plays, Anya takes the adjacent star, which is in an even position. Note that the sum of the values of the stars in the odd positions is

$$1 + 2 + \cdots + (m+1) = \frac{(m+1)(m+2)}{2},$$

and the sum of the values of the stars in the even positions is

$$(m+2) + (m+3) + \cdots + (2m+1) = \frac{3m}{2}(m+1).$$

Since $m \geq 2$, we have

$$\frac{3m}{2}(m+1) > \frac{(m+1)(m+2)}{2},$$

so Anya wins.

Assume that $n = 4m + 4$ with $m \geq 1$. Then Anya uses a similar strategy to force a draw or a win for Anya. First, Anya places the stars with values $m+1, m+2, \ldots, 2m$ in the first m odd positions from left to right, the stars with values $2m+1, 2m+2, \ldots, 3m$ in the next m odd positions from left to right, the stars with values $3m+1, 3m+2, \ldots, 4m$ in the first m even positions from left to right, and the stars with values $1, 2, \ldots, m$ in the next m even positions from left to right. In other words, Anya places the stars in the order

$$\begin{aligned} &m+1, 3m+1, m+2, 3m+2, m+3, 3m+3, \ldots, 2m, 4m, \\ &2m+1, 1, 2m+2, 2, \ldots, 3m, m \end{aligned}$$

from left to right. Anya play by taking the star adjacent to the star taken by Becky in each turn.

Claim — As a consequence of the above strategy, the game ends in a draw or a win for Anya.

Proof of the Claim. Let us first consider the case when Becky takes more or an equal number of stars from the left end of the row than from the right end of the row. Note that Anya takes the stars with values $3m+1, 3m+2, \ldots, 4m$. As per Anya's strategy,

$$\begin{aligned} &\text{the sum of the values of stars taken by Anya} \\ &\quad - \text{the sum of the values of stars taken by Becky} \\ &\geq (3m+1) + (3m+2) + \cdots + (4m) \\ &\quad - ((m+1) + (m+2) + \cdots + (2m)) \end{aligned}$$

$$
\begin{aligned}
&\quad - |(2m+1)-1| - |(2m+2)-2| - \cdots - |3m-m| \\
&= 2m^2 - 2m^2 \\
&= 0.
\end{aligned}
$$

Next, we consider the case when Becky takes more stars from the right end of the row than from the left end of the row. Note that Anya takes the stars with values $2m+1, 2m+2, \ldots, 3m$. As per Anya's strategy,

$$
\begin{aligned}
&\text{the sum of the values of stars taken by Anya} \\
&\quad - \text{the sum of the values of stars taken by Becky} \\
&\geq (2m+1) + (2m+2) + \cdots + (3m) \\
&\quad - (1 + 2 + \cdots + m) \\
&\quad - |(3m+1) - (m+1)| - |(3m+2) - (m+2)| - \cdots - |4m - 2m| \\
&= 2m^2 - 2m^2 \\
&= 0.
\end{aligned}
$$

This proves the claim. □

This proves that Becky has a winning strategy if and only if $n \equiv 2 \pmod 4$. ■

Exercise 5 (All-Russian Mathematical Olympiad 2025 Grade 10 Day 2 P5, , also 11.5, proposed by I. I. Bogdanov)**.** Let n be a natural number. The numbers $1, 2, \ldots, n$ are written in a row in some order. For each pair of adjacent numbers, their greatest common divisor is calculated and written on a sheet. What is the maximum possible number of distinct values among the $n-1$ integers obtained?

> **Walkthrough** —
>
> **(a)**

Solution 5. In the following, we assume that $n \neq 1$. Using division algorithm, for any two distinct positive integers a and b, it follows that

$$
\gcd(a,b) \leq \begin{cases} a \leq \frac{b}{2} & \text{if } a \text{ divides } b, \\ \frac{a}{2} & \text{if } a \text{ does not divide } b, \end{cases}
$$

and hence $\gcd(a,b) \leq \frac{1}{2}\max(a,b)$. This implies that the GCD of any two adjacent numbers in the row is at most $n/2$. Therefore, the maximum possible number of distinct GCDs obtained, is at most $\lfloor n/2 \rfloor$.

We will show that this upper bound can be achieved by the following arrangement of the numbers $1, 2, \ldots, n$. For each odd positive integer $k \leq n$,

write the integers between 1 and n in a block $\mathcal{B}_k$, having k as its largest odd divisor, in an increasing order, that is, the elements of $\mathcal{B}_k$ are put in the order

$$k, 2k, 4k, \ldots, 2^{e_k}k,$$

where e_k is the largest non-negative integer such that $2^{e_k}k \leq n$. Next, we put the blocks $\mathcal{B}_k$ in a row as k increases across the odd positive integers less than or equal to n.

Claim — Let $1 \leq m \leq n/2$ be a positive integer. Then m is equal to the GCD of some pair of adjacent numbers in the above arrangement.

Proof of the Claim. Since m is at most $n/2$, it follows that $2m$ is at most n. If k denotes the largest odd divisor of m, then k is less than or equal to m and hence $k \leq n$. Hence, in the above arrangement, the block $\mathcal{B}_k$ has appeared, and the numbers m and $2m$ are adjacent in this block. Note that the GCD of m and $2m$ is m. This proves the claim. □

Therefore, the GCDs of the adjacent numbers in the above arrangement include all the integers $1, 2, \ldots, \lfloor n/2 \rfloor$. Since the number of distinct GCDs obtained is at most $\lfloor n/2 \rfloor$, it follows that the maximum possible number of distinct GCDs obtained is $\lfloor n/2 \rfloor$. ■

Remark. One may also observe that if ℓ denotes the largest GCD obtained from the adjacent numbers in the row, then it is the gcd of two distinct integers between 1 and n, and the larger of these two integers is at least 2ℓ, implying that $\ell \leq n/2$.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)